Michael Bleher, Dr.

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Academic Positions

2023–present **Postdoctoral Researcher**, *Heidelberg University*.

STRUCTURES Cluster of Excellence
Exploratory Project: New Methods for Single Cell Data Analysis – a Promising Future Direction for STRUCTURES?
Prof. Dr. Fred Hamprecht, Prof. Dr. Anna Marciniak-Czochra, Prof. Dr. Anna Wienhard.

ERC Project 'Perpetuating Stemness: From single-cell analysis to mechanistic spatio-temporal models of neural stem cell dynamics'.
Prof. Dr. Anna Marciniak-Czochra.

Research Directions:

- Geometry-preserving low-dimensional representations of scRNA-seq data.
- Topological signatures for model selection in neuron differentiation dynamics.
- Topology of spatial transcriptome organisation.

Education

2017–2023 **PhD Physics and Mathematics (Interdisciplinary)**, *Heidelberg University*.
Advisor: Prof. Dr. Johannes Walcher
'Haydys-Witten Instantons and Symplectic Khovanov Homology'

2013–2016 **M.Sc. Physics**, *Heidelberg University*.
Advisor: Prof. Dr. Johannes Walcher
'A Survey of Defects in $\mathcal{N} = 4$ Supersymmetric Yang-Mills Theory'

2014–2015 **Graduate Course**, *University of Durham, UK* (ERASMUS+ Exchange).
M.Sc. in Particles, Strings and Cosmology at the Centre for Particle Theory.

2010–2013 **B.Sc. Physics**, *Heidelberg University*.
Advisor: Dr. Werner Rodejohann
'Neutrinoloser Doppelbeta-Zerfall – Untersuchung einer Methode zur Auswahl eines Nuklearen Matrix-Elements'

Grants and Fellowships

2025 **Workshop Funding**, *Cluster of Excellence STRUCTURES, Heidelberg University*.
Funding for the Workshop on Geometry, Topology, and Machine Learning, held jointly by MPI-MiS and STRUCTURES. EUR 20,000, with F. Jensen and L. Maier.

2025 **DAAD Conference Grant**, *German Academic Exchange Service (DAAD)*.
Support for attending the 'Geometric Realization of the AATRN', Chicago.

2017–2020 **Distinguished Doctoral Fellowship**, *Heidelberg Graduate School of Fundamental Physics*.

Publications

4. **Bleher**, **M.** (2026). *The Tangled Web They Weave: Exploring Neural Representations with Directed Simplicial Complexes*. arXiv:2611.00786 [cs.LG] (accepted at NeurReps 2026, proceedings track).
3. **Bleher**, **M.**, Jensen, F., Maier, L., Taha, D. and Wienhard, A. (2026). *Preface of Geometry, Topology, and Machine Learning 2025*. In: Proceedings of the Geometry, Topology, and Machine Learning Workshop, PMLR 325, pp. i–vii.
2. **Bleher**, **M.** (2026). *A Family of Instanton-Invariants for Four-Manifolds and Their Relation to Khovanov Homology*. SIGMA 22, 028, 47 pages. DOI: 10.3842/SIGMA.2026.028. arXiv:2412.13285 [math-ph].
1. **Bleher**, **M.** (2025). *Adiabatic Solutions of the Haydys-Witten Equations and Symplectic Khovanov Homology*. Advances in Theoretical and Mathematical Physics, accepted. arXiv:2501.01365 [math-ph].

Preprints

9. **Bleher**, **M.**, Gómez, M., Mémoli, F., Neumann, M. and Ott, A. (2026). *Distributions of metric n-point invariants and applications to molecular evolution*. arXiv:2610.26070 [math.MG].
8. **Bleher**, **M.**, Brazke, D. and Nill, S. (2026). *A Riemannian Autocorrelation Function and its Application to Non-Local Isoperimetric Energies*. arXiv:2601.10481 [math.AP].
7. De Bodt, C., Diaz-Papkovich, A., **Bleher**, **M.**, Bunte, K., Coupette, C., Damrich, S., Sanmartin, E. F., Hamprecht, F. A., Horvát, E.-Á., Kohli, D., Krishnaswamy, S., Lee, J. A., Lelieveldt, B. P. F., McInnes, L., Nabney, I. T., Noichl, M., Poličar, P. G., Rieck, B., Wolf, G., Mishne, G. and Kobak, D. (2025). *Low-Dimensional Embeddings of High-Dimensional Data*. arXiv:2508.15929 [cs.LG].
6. Bleher, J. and **Bleher**, **M.** (2024). *An Algebraic Framework for the Modeling of Limit Order Books*. arXiv:2406.04969 [q-fin.TR].
5. **Bleher**, **M.** (2023). *The Decoupled Haydys-Witten Equations and a Weitzenböck Formula*. arXiv:2307.15056 [math.DG].
4. **Bleher**, **M.** (2023). *Growth of the Higgs Field for Kapustin-Witten Solutions on ALE and ALF Gravitational Instantons*. arXiv:2306.17017 [math.DG].
3. Neumann, M., **Bleher**, **M.**, Hahn, L., Braun, S., Obermaier, H., Soysal, M., Caspart, R. and Ott, A. (2022). *MuRiT: Efficient Computation of Pathwise Persistence Barcodes in Multi-Filtered Flag Complexes via Vietoris-Rips Transformations*. arXiv:2207.03394 [math.AT].
2. **Bleher**, **M.**, Hahn, L., Patino-Galindo, J. A., Carriere, M., Bauer, U., Rabadan, R. and Ott, A. (2021). *Topology Identifies Emerging Adaptive Mutations in SARS-CoV-2*. arXiv:2106.07292 [q-bio.PE].
1. Bleher, J., **Bleher**, **M.** and Dimpfl, T. (2020). *From Orders to Prices: A Stochastic Description of the Limit Order Book to Forecast Intraday Returns*. arXiv:2004.11953 [q-fin.TR].

Presentations

21. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. Division of Differential Equations and Applications of Mathematics Seminar, Gdansk University of Technology, Gdansk, Poland (invited talk). 28th May 2026.
20. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. MATRIX Workshop: Applications of Computational Geometry and Topology to Data Analysis of Complex Biological Data, Creswick, Australia (invited talk). 14th May 2026.
19. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. EPFL Topology Seminar, Lausanne, Switzerland (invited talk). 2nd Apr. 2026.

18. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. ATMCS 12, Leipzig, Germany (flash talk). 26th Jan. 2026.
17. *Population Dynamics, scRNA-seq, and Euler Characteristic Profiles*. Velten Lab Seminar, Heidelberg, Germany (invited talk). 20th Jan. 2026.
16. *Topological Signatures of Convergence in Viral Evolution*. Dioscuri TDA Seminar, Warsaw, Poland (invited talk). 12th Jan. 2026.
15. *Population Dynamics, scRNA-seq Data, and the Euler Characteristic Profile*. PEPS Seminar, Heidelberg, Germany. 30th Oct. 2025.
14. *Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. Rutgers Gauge Theory Seminar (invited talk). 14th Oct. 2025.
13. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. Geometric Realization of the AATRN, Chicago, USA (poster). 19th Aug. 2025.
12. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. Young Topologists Meeting 2025, Stockholm, Sweden (contributed talk). 25th June 2025.
11. *Persistence and Coarse-Graining in Dynamical Biological Systems*. TDA Symposium at MPI-MCG, Dresden, Germany (invited talk). 12th Dec. 2024.
10. *Fast Computation of Pathwise Persistence in Pandemic-Scale SARS-CoV-2 Genome Data*. 4th Workshop on Computational Persistence, TU Graz, Graz, Austria. 23rd Sept. 2024.
9. *Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. Gauge Theory and Mathematical Physics Seminar, Morningside Center of Mathematics, Beijing, China (invited talk). 3rd July 2024.
8. *RNA Velocity Embeddings in Curved Spaces - Exploring Cellular Dynamics*. Seminar 24122, Dagstuhl, Germany. 20th Mar. 2024.
7. *On Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. HU Gauge Theory Research Seminar, Berlin, Germany (invited talk). 31st Jan. 2024.
6. *Haydys-Witten Instantons in the Gauge Theoretic Approach to Khovanov Homology*. ULB Geometry Seminar, Brussels, Belgium (invited talk). 4th Dec. 2023.
5. *Topological Signatures of Convergence in Viral Evolution*. CompTopNN Meeting 2023, Seville, Spain (invited talk). 8th Nov. 2023.
4. *Feature Representation of scRNA Data in Symmetric Spaces*. STRUCTURES Symposium, Heidelberg, Germany (poster). 20th July 2023.
3. *Persistent Homology Detects Emerging Adaptive Mutations*. TDA Journal Club, Heidelberg, Germany. 7th June 2021.
2. *Welcome Notes and an Introduction to Mapper*. Heidelberg TDA Workshop 2020, Heidelberg, Germany (organiser). 26th Oct. 2020.
1. *Nahm Pole Boundary Conditions and Lagrangian Intersection Floer Homology*. Quantum Fields, Knots, and Strings, Warsaw, Poland (poster). 2018.

Academic Service

Editorial Work

2026 **PMLR Volume 325: *Geometry, Topology, and Machine Learning***, *Proceedings of Machine Learning Research*.
Editor, with F. Jensen, L. Maier, D. Taha, A. Wienhard.

Workshops and Seminars

2025 **Workshop on Geometry, Topology, and Machine Learning**, *MPI-MiS, Leipzig*.
Organiser, with F. Jensen, L. Maier, D. Taha, A. Wienhard.

2025–PRESENT **Persistent Seminar**, *TDA Research Seminar, Heidelberg & MPI of Biochemistry*.
Organiser, with F. Jensen (Heidelberg) and B. Stolz-Pretzer (MPIB).

2024–PRESENT **Tea, Coffee, Cake & TDA**, *Invited Speaker Series, Heidelberg University*.
Organiser, with F. Jensen.

2023 **Workshop on Topological Methods in Data Analysis**, *Heidelberg University*.
Organiser, with F. Jensen, V. Noel, A. Marciniak-Czochra.

2020, 2021 **Workshop on Topological Methods in Data Analysis**, *Heidelberg University*.
Organiser, with D. Spitz, M. Schmahl, A. Wienhard.

2019–2023 **Topics in TDA**, *Journal Club, Heidelberg University*.
Organiser, with F. Jensen, V. Noel, D. Spitz, M. Schmahl, A. Wienhard.

Peer Review

NeurReps 2026 (NeurIPS workshop), Journal of Computational Biology.

SOFTWARE

N-INVARIANTS **Metric *n*-point Invariants**, *Python package*.
Sampling distributions of metric n-point invariants: code and results for *Distributions of metric n-point invariants and applications to molecular evolution* (2026).
github.com/subthaumic/n-invariants, doi: 10.5281/zenodo.23210358

NEURAL-DSC **Neural Directed Simplicial Complexes**, *Python package*.
Filtered directed simplicial complexes of neural networks: code and results for *The Tangled Web They Weave* (NeurReps 2026).
github.com/subthaumic/neural-dsc, doi: 10.5281/zenodo.23777983

CHROMABENCH **Chromatic TDA Benchmark**, *Python package*.
A family of two-colour point clouds for checking whether a chromatic topological-data-analysis method can distinguish between shape and colour distributions.
github.com/subthaumic/chromabench, doi: 10.5281/zenodo.22772348

GNE **Geometric Neighbour Embeddings**, *Python package*.
A dimensionality reduction technique similar to t-SNE and UMAP that preserves the local geometric relationships between data points.
github.com/subthaumic/gne

MARKOVMODUS **Markov-modulated Splicing Model**, *Python package*.
Generates unspliced/spliced RNA count data for benchmarking trajectory inference and RNA velocity methods.
github.com/subthaumic/markovmodus

MURIT **Multiparameter Rips Transform**, *Go application*.
A Ripser add-on enabling exploration of multi-persistence of filtered metric spaces.
github.com/tdalife/murit

EVOTREC **Topological Recurrence in Evolution**, *Python package*.
Topological recurrence analysis of genome alignments using topological data analysis.
github.com/ottamj/evotrec

Teaching

2022 **Mathematics Refresher for Master Students in Economics**, *Teaching Assistant, Prof. Dr. Thomas Dimpfl, University of Hohenheim* (available on YouTube).

2020–2021 **Helpdesk for Math Students**, *Teaching Assistant, Heidelberg University*.

2019 **Theoretische Physik IV**, *Teaching Assistant, Prof. Dr. Arthur Hebecker, Heidelberg University*.

2018 **Theoretische Physik I**, *Teaching Assistant, Prof. Dr. Tilman Plehn, Heidelberg University*.

2018 **Höhere Mathematik für Physiker III**, *Teaching Assistant, Prof. Dr. Hans Knüpfer, Heidelberg University*.

2018 **Theoretische Physik II**, *Teaching Assistant, Prof. Dr. Matthias Bartelmann, Heidelberg University*.

2017 **Applied Topology I**, *Teaching Assistant, Prof. Dr. Daniel Roggenkamp, University of Mannheim*.